PRISMSPACE - INTELLIGENT MULTI-AGENT PLATFORM



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Project Group 01

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01. Introduction: PrismSpace Web Platform

PrismSpace is an interactive, **web-based multi-agent intelligence platform** and workspace website that allows users to orchestrate autonomous AI swarms directly from modern web browsers. It converts natural-language user goals into dependency-aware **Directed Acyclic Graphs (DAGs)** for parallel sub-task execution. Built as a responsive web application (Next.js 15, React 19) accessible from any browser with zero local setup, it features an interactive visual DAG swarm visualizer with real-time token and step streaming. The platform pairs Groq, OpenAI, and NVIDIA NIM with standardized tool connectivity via Model Context Protocol (MCP), while storing all session state locally in client-side IndexedDB (`PrismDB`) with human-in-the-loop approval gates for privacy and safety.

02. Problem Statement: Core Challenges in AI Agent Systems

1. **Monolithic LLM Bottleneck & High Cost:** Traditional platforms route all planning, reasoning, and tool use through a single massive LLM, causing severe prompt bloat, high latency, and $3\times$ higher inference expenses.
   ➔ **PrismSpace Solution:** Dynamically routes sub-tasks to cost- and latency-optimal models (Groq for rapid inference, OpenAI/NIM for complex reasoning).

2. **Sequential & Rigid Task Execution:** Existing agent chains run strictly in linear sequence; independent sub-tasks wait needlessly in line, and a single step crash causes total pipeline collapse.
   ➔ **PrismSpace Solution:** Deconstructs goals into dynamic Directed Acyclic Graphs (DAGs) to execute independent tasks in parallel ($K \leq 3$) with isolated fault recovery.

3. **Fragmented Tool Silos & Black-Box Runs:** Proprietary integrations isolate essential productivity tools (Gmail, Drive, Terminal, Local Files), while users remain blind during long runs with no status logs.
   ➔ **PrismSpace Solution:** Standardizes all tool connections via Model Context Protocol (MCP) paired with real-time execution telemetry and progress logs.

4. **Safety Hazards & Cloud Data Lock-in:** Autonomous agents execute irreversible actions (sending emails, modifying files) without oversight, while cloud databases expose private user data.
   ➔ **PrismSpace Solution:** Enforces Human-in-the-Loop (HITL) approval gates, ORPO alignment, and zero-latency local browser database storage (`PrismDB`).

03. Comparative Analysis: Existing Platforms vs. PrismSpace

Architectural Benchmark: Evaluating PrismSpace against industry-standard conversational AI, autonomous agents, and swarm frameworks.

Capability Dimension	ChatGPT Web (OpenAI)	AutoGPT (Autonomous Loop)	CrewAI (Multi-Agent Swarm)	PrismSpace (Ours) Local Multi-Agent Platform
Execution Topology	Linear single-thread prompt dialog	Unbounded recursive loop (risk of infinite loops)	Sequential or static hierarchical agent chains	**Dynamic DAG Scheduling:** Parallel execution ($K \leq 3$, join barriers)
Model Strategy & Routing	Vendor locked (single OpenAI model per session)	Single model API backend (typically GPT-4)	Statically configured per agent persona	**Dynamic Multi-Model Router:** Routes Groq, OpenAI, NIM by latency/cost
Tool Connectivity Protocol	Proprietary OpenAI Plugins & Function Calls	Custom Python scripts & local shell commands	LangChain tool wrappers (rigid contracts)	**Model Context Protocol (MCP):** Standardized servers (Gmail, Drive, FS, Web)
Observability & Streaming	Text token stream only (internal steps opaque)	Terminal CLI logs only; zero visual graph	Terminal text traces; server-side execution	**Live SSE Stream + Visual DAG:** Real-time node state and telemetry
Safety & Quality Control	Closed cloud content filter; no user approval gate	Fixed step budgets; no output verification	Agent self-critique (incurs token overhead)	**HITL Approval Gates + ORPO:** High-risk tasks paused; preference aligned
Persistence & Privacy	Remote cloud storage (privacy & latency exposure)	Local filesystem artifacts; no structured database	Cloud DB or ephemeral in-memory state	**Local-First PrismDB:** Zero-latency browser IndexedDB (`useLiveQuery()`)

While ChatGPT is optimized for single-model dialog and frameworks like CrewAI/AutoGPT for Python-scripted execution, **PrismSpace** uniquely unifies **dynamic DAG planning**, **open MCP tools**, **multi-model routing**, and **local-first privacy** in a reactive workspace.

04. Project Objectives

1. **Context-Aware Task Planning:** Convert everyday user goals (communication, file management, daily schedules) and technical requests into structured Directed Acyclic Graphs (DAGs) for organized, dependency-safe execution.

2. **Intelligent Model & Agent Routing:** Dynamically route each sub-task to the most suitable Large Language Model (LLM) backend and dispatch specialized, context-aware agents in parallel.

3. **Practical Tool Execution via MCP:** Connect agents to everyday productivity tools and development utilities (Gmail, Google Drive, Filesystem, Terminal CLI, GitHub) through the Model Context Protocol (MCP), accompanied by live Server-Sent Events (SSE) updates.

4. **Safety Verification & Local Persistence:** Validate outputs with human approval checkpoints and Odds Ratio Preference Optimization (ORPO), storing results reactively in local IndexedDB (Indexed Database: `PrismDB`).

05. PrismSpace Workflow (Figure 1.0)

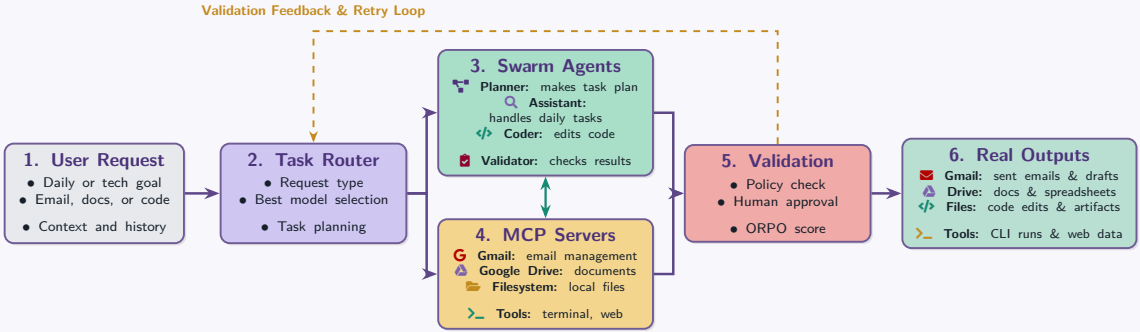


Figure 1.0: PrismSpace Workflow | **DAG** = Directed Acyclic Graph | **MCP** = Model Context Protocol | **ORPO** = Odds Ratio Preference Optimization | **SSE** = Server-Sent Events

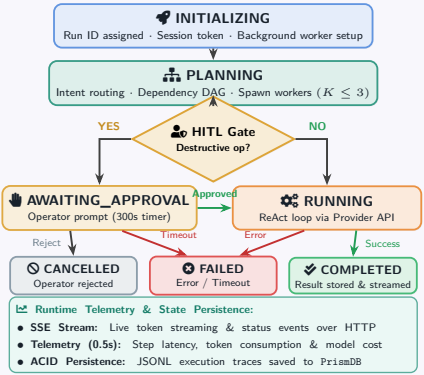
06. System Design Principles

- **Layered Decoupling:** Separates the React client UI, FastAPI bridge, and foundation LLMs (each layer has one clear role).
- **Dynamic DAG Scheduling:** Runs dependent sub-tasks in parallel ($K \leq 3$; additional tasks are queued) for daily and technical workflows.
- **Standardized Protocol (MCP):** Uses open JSON-RPC tool contracts (a common way for agents and tools to communicate).
- **Human-in-the-Loop:** Sensitive operations require explicit user approval (a safety checkpoint before execution).
- **Local IndexedDB (`PrismDB`):** ACID browser storage (reliable local data) for sessions, tasks, tool history, and files.
- **Reactive Subscriptions:** Dexie `useLiveQuery()` updates the UI automatically when data changes.
- **Server-Sent Events (SSE):** Streams tokens and agent telemetry over persistent HTTP (live one-way updates).
- **Calibrated Preference Gating:** ORPO ranks and filters outputs to improve quality and safety.
- **Design Philosophy:** Context-aware, sandboxed, and observable multi-agent workspace (safe and traceable by design).

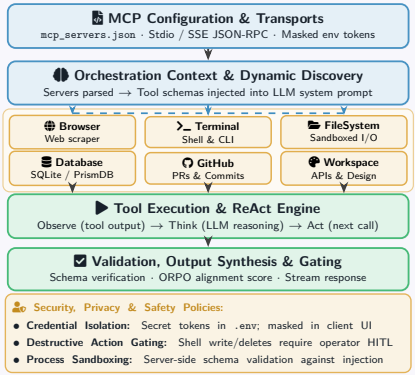
07. Agent Lifecycle, MCP Integration & Runtime Flow in PrismSpace

Runtime Orchestration & Tool Architecture: State-machine lifecycle with Human-in-the-Loop (HITL) safety gates, bi-directional MCP tool discovery, and real-time Server-Sent Events (SSE) telemetry.

Agent Lifecycle & State Flow



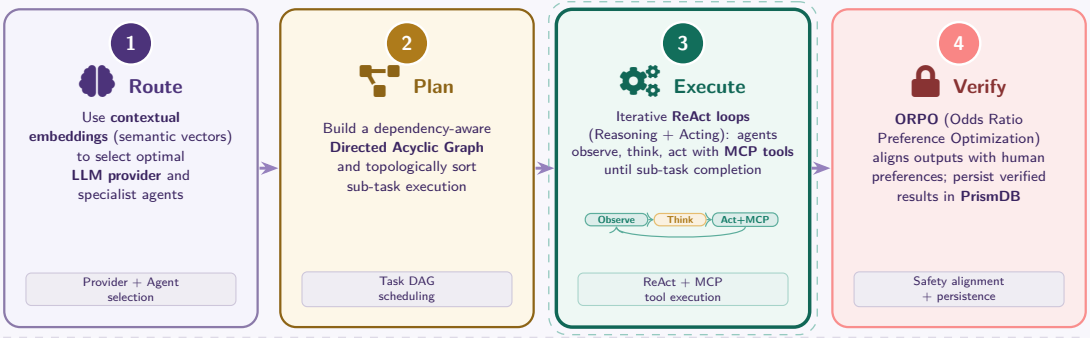
MCP Tool Integration & Runtime Flow



Core Infrastructure: **FastAPI Async Engine** (`hive_api.py`) | **Next.js 14 Frontend** (`AgentSwarm.tsx`) | **Dexie PrismDB** (ACID Storage) | **Multi-LLM Routing** (Groq, NVIDIA, OpenAI)

08. Proposed Methods

PrismSpace Orchestration Pipeline: An end-to-end multi-agent architecture designed to transform natural language requests into reliable execution traces. It leverages contextual embeddings for intelligent routing, DAG-based task scheduling, iterative ReAct (Reasoning + Acting) loops via MCP tools, and validates outputs through ORPO (Odds Ratio Preference Optimization) alignment with human verification gates.



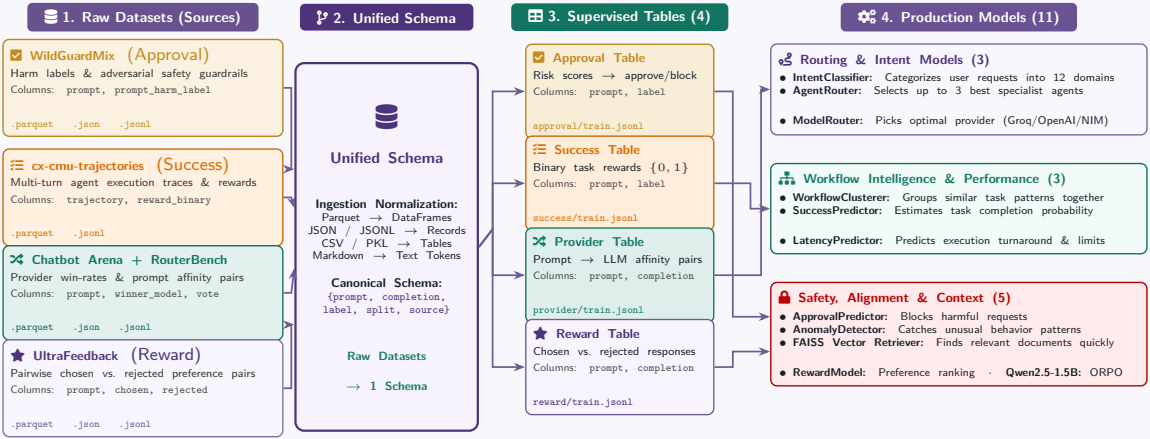
09. Model Responsibilities & Inference Flow

Intent: The **underlying goal** of a user's request. Determines *task type* (research, coding, tool use, synthesis) for routing to agents & tools.

1. Intent Classifier	**Prompt** → **Extracted features** → **classify intent** (research, tool use, coding, everyday synthesis)	**Task type** `intent_classifier.joblib`
2. Agent Router	**Intent + task features** → predict **multiple agent labels** → form the **specialist team**	**Agent selection** `agent_router.joblib`
3. Workflow Planner	**Workflow templates** → estimate **completion probability** → choose a **dependency-safe DAG plan**	**Plan reuse + success** `workflow_templates.pkl` `workflow_success_predictor.joblib`
4. Provider Routing	**Candidate task** → choose **provider/model** → compare **predicted latency** and **token cost**	**Provider + cost balance** `model_router.joblib` `latency_predictor.joblib` `cost_predictor.joblib`
5. Safety Gate	**Planned action** → score **approval risk** + **detect anomalies** → **approve, pause, or block** execution	**Risk approval & safety** `approval_predictor.joblib` `anomaly_detector.joblib`
6. Context & Quality	**Query** → **FAISS semantic retrieval** → add **context** → generate & **ORPO-align response**	**Evidence + output quality** `faiss.index` · `faiss.vectorizer.joblib` `reward_orpo_v2/adapter_model.safetensors`

10. Unified Dataset Preparation & Multi-Model Training Pipeline

Raw multi-source datasets across multiple formats are normalized into a **unified schema**, curated into **4 supervised tables**, and trained into **11 production models**.



11. How the System Works: From Request to Result

Stage 1: Ingestion & Features

- **Capture request:** Parse user input (text/voice/API).
- **Clean & normalize:** Remove noise, handle missing data with NLP preprocessing.
- **Extract features:** Tokenize into semantic embeddings using transformer models.

Stage 2: Intelligence & DAG

- **Classify intent:** ML classifier identifies task type from 12 categories (99.6% accuracy).
- **Choose agents & models:** Route sub-tasks to specialized LLM agents.
- **Build the DAG:** Create dependency graph ensuring safe execution order.

Stage 3: Execution & Storage

- **Execute:** Run up to 3 parallel tasks using Model Context Protocol (MCP) tools.
- **Track progress:** Real-time telemetry, node state transitions, and execution logs.
- **Verify & store:** ORPO quality check, persist results to PrismDB.

Building & Executing the Task Graph $\mathcal{G}(V, E)$

1. User Goal Prompt
"Analyze news+CSV, email summary"

→

2. Split into Vertices V
Modular async tasks A, B, C, D

→

3. Connect Edges E
Map output return → input args

→

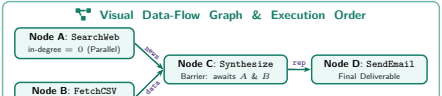
4. Topological Run
Run in-degree $= 0 \implies$ join barrier

Computer Science Explanation (By Concrete Example):

- **Vertices (V) = Sub-Task Functions (Async Call Units):**
  Node A: `news = await SearchWeb("Tesla news")` *[Researcher]*
  Node B: `data = await FetchCSV("metrics.csv")` *[Data Agent]*
  Node C: `rep = await Synthesize(news, data)` *[Writer Agent]*
  Node D: `await SendEmail(to="lead", body=rep)` *[Email Tool]*
- **Edges (E) = Data Pipes & Synchronization Barriers:**
  Edge $\langle A, C \rangle$ pipes return value `news`; edge $\langle B, C \rangle$ pipes `data`.
  Barrier: task C blocks until both promises A & B resolve.
- **Connection Basis (Input ← Output Matching):**
  Function C links to A & B because C's parameters require their outputs. A and B share no edge because neither depends on the other's data.
- **Execution Schedule (Topological Concurrency):**
  Tasks with in-degree $= 0$ (A, B) execute **concurrently in parallel**.

When A & B complete, C's in-degree drops to $0 \implies C$ runs $\implies D$ runs.

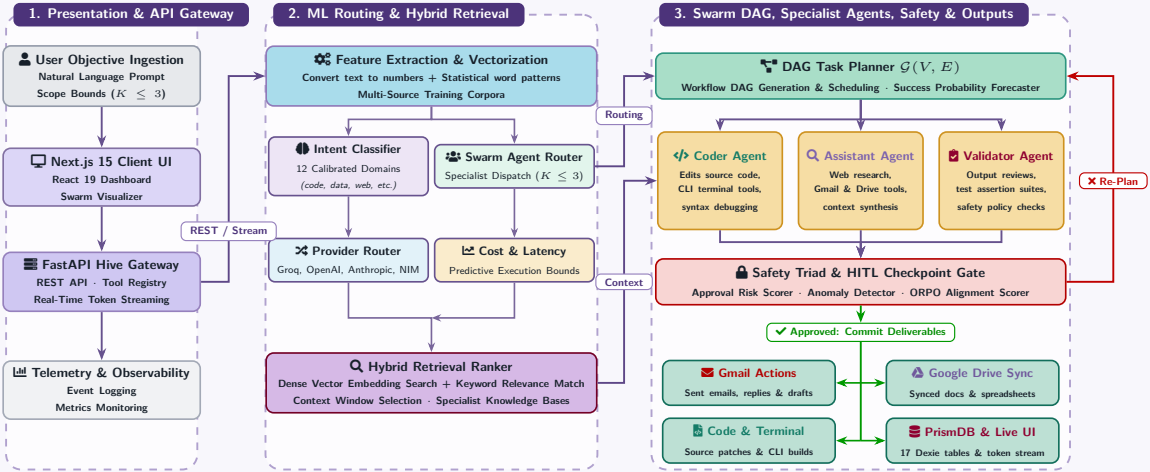
Visual Data-Flow Graph & Execution Order



Parallel Thread 1: Worker 1 executes Node A → returns `news`
Parallel Thread 2: Worker 2 executes Node B → returns `data`
Join Barrier: Node C consumes `news` + `data` → outputs `rep`

Serial Finish: Node D consumes `rep` → sends email

12. PrismSpace System Architecture & Multi-Model Swarm Workflow (Figure 1.2)



13. Flowchart: Multi-Agent Execution Flow (Figure 1.3)

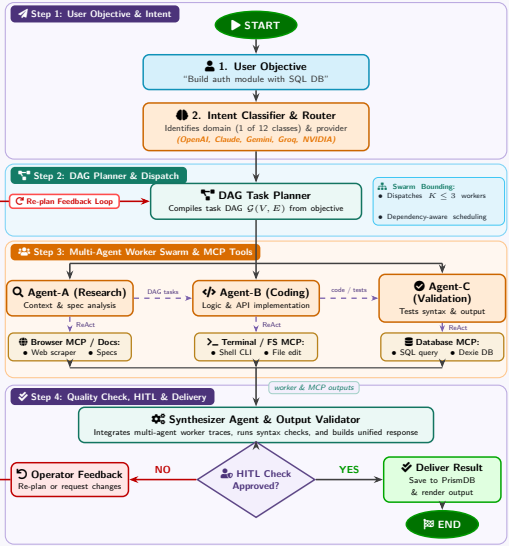


Figure 1.3: Simplified Multi-Agent Communication Flow in PrismSpace

14. Browser Database Schema: PrismDB (Figure 1.4)

agent_chat_sessions

&id	string (UUID, PK)
title	string
updatedAt	number (Index)
createdAt	number (Index)

agent_chat_messages

++id	number (Auto-Inc PK)
sessionId	string (FK → sessions.id)
[sessionId+createdAt]	Compound Index
agentId	string (Index)
role	'user' | 'assistant' | 'tool'
status	'pending'|'streaming'|'done'
content	string (Text)
createdAt	number (Index)

ai_tool_history

++id	number (Auto-Inc PK)
sessionId	string (FK → sessions.id)
toolType	string (bash|web|code)
input	text (Args / JSON Payload)
output	text (Logs / Return Value)
createdAt	number (Index)

Agent Key-Value Stores

&key settings	JSON (API keys, model config)
&key agent_state	Active Swarm DAG
&key user_profile	JSON (System Instructions)
&key files	Blob / MimeType (Deliverables)

Figure 1.4: PrismDB - AI-Agent Browser IndexedDB Schema (Dexie.js v3) | **PK**: Primary Key **FK**: Foreign Key **Index**: B-tree index
sessions.id (Shared FK) → messages.sessionId & tool_history.sessionId | useLiveQuery() Reactive UI | ACID Dexie.js

Algorithm: Step 1 – Understand the User Prompt

Step 1: Prompt Parsing and Structured Representation

1.1 Ingest raw natural language prompt $\mathcal{T}_{\text{user}}$ via client interface and extract core objective.

1.2 Retrieve persistent session context and workspace history from browser IndexedDB (`PrismDB`):

- **1.2.1** Load prior interaction turns, agent memory checkpoints, and active user configurations.
- **1.2.2** Resolve local file system references, target project repositories, and access tokens.

1.3 Construct strongly-typed operational execution tuple:

$$T = \langle \text{Goal, Constraints, Context, } \mathcal{M}_{\text{tools}}, \Phi_{\text{out}} \rangle$$

Where: Goal: Target objective | Constraints: Safety & SLA bounds | Context: Session state in `PrismDB` | $\mathcal{M}_{\text{tools}}$: Authorized MCP tools | Φ_{out}: Deliverable schema contract.

- **1.3.1** Parse explicit operational constraints (deadlines, schema specifications, output formats).
- **1.3.2** Bind authorized Model Context Protocol (MCP) tool capabilities and environment permissions.

1.4 Perform input boundary and security screening:

- **1.4.1** Evaluate prompt against token limits and scan for prompt injection patterns.
- **1.4.2** Flag state-mutating actions requiring mandatory human-in-the-loop authorization.

Step 2: Intent Classification and Capability Check

2.1 Categorize user prompt into functional operational domains:

$$\text{Domain} \in \{\text{Productivity/Automation, Document Synthesis, Web Research, Code Engineering}\}$$

Where: Productivity: Communication & planning | Synthesis: Summarization | Research: Web search & scraping | Coding: CLI & code synthesis.

- **2.1.1** Compute dense semantic embeddings and syntactic keyword vectors for input prompt.
- **2.1.2** Route through multi-class classifier head $\hat{\mathbf{y}}_{\text{intent}}$ to determine target capability domain.

2.2 Profile required system capabilities and operational risk levels:

- **2.2.1** Determine required subsystem resources (live web search, local filesystem I/O, terminal execution).
- **2.2.2** Enforce safety categorization: Read-Only (autonomous execution) vs. State-Mutating (approval-gated).

2.3 Select optimal execution pathway:

- **2.3.1** Route single-step queries to direct, low-latency single-model inference pipeline.
- **2.3.2** Dispatch multi-step complex requests to DAG Planner agent for task decomposition.

Step 3: DAG Decomposition and Model Routing

3.1 Deconstruct objective into Directed Acyclic Graph $\mathcal{G} = (V, E)$:

- **3.1.1** Instantiate discrete task vertices $v_i \in V$ with atomic execution contracts and output targets.
- **3.1.2** **Data Dependencies as Directed Edges** ($e = \langle v_i, v_j \rangle \in E \iff \mathcal{O}_i \cap \mathcal{I}_j \neq \emptyset$): Task v_j requires input artifacts produced by v_i (e.g., synthesis requires scraped research).
- **3.1.3** **Execution Barrier:** Consumer v_j remains blocked until producer v_i delivers $\mathcal{O}_i$, guaranteeing tasks never run on missing or partial data.

3.2 Compute topological task execution schedule $\pi = \mathrm{TopologicalSort}(\mathcal{G})$:

- **3.2.1** Identify independent vertices (in-degree $= 0$) for concurrent parallel dispatch ($K \leq 3$).
- **3.2.2** Schedule downstream vertices dynamically upon upstream output event emission.

3.3 Solve multi-objective provider routing optimization for each sub-task v_i:

$$\text{Provider}^* = \arg\min_m \left(\lambda_{\text{cost}} \, \text{Cost}_m + \lambda_{\text{latency}} \, \text{Latency}_m \right) \quad \text{s.t. } \text{Quality}_m \geq \tau$$

Where: Provider*: Optimal LLM backend | m: Candidate model | $\text{Cost}_m, \text{Latency}_m$: Normalized price & delay | $\lambda_{\text{cost}}, \lambda_{\text{latency}}$: Balancing weights | τ: Quality barrier.

- **3.3.1** Route high-throughput, latency-critical nodes to LPU backends (Groq / Llama-3).
- **3.3.2** Route deep reasoning and code generation tasks to frontier backends (OpenAI GPT-4o, Claude 3.5).

Step 4: Role-Based Task Assignment and Dispatching

4.1 Query specialist agent registry and capability profiles:

$$\mathcal{A} = \{\text{Planner, Assistant, Researcher, Coder, Validator}\}$$

4.1.1 Evaluate persona system prompts, specialized instruction boundaries, and assigned MCP tools.

4.1.2 Verify agent worker availability and monitor concurrent thread utilization.

4.2 Execute capability matching algorithm for each sub-task node v_i:

$$\text{Agent}^* = \arg\max_{a \in \mathcal{A}} \ \text{CosineSimilarity}\big(\mathbf{Req}(v_i),\, \mathbf{Cap}(a)\big)$$

Where: Agent*: Assigned specialist persona | $\mathcal{A}$: Swarm agent registry | $\mathbf{Req}(v_i)$: Task requirement vector | $\mathbf{Cap}(a)$: Agent capability embedding | CosineSimilarity: Semantic alignment.

4.2.1 Match information gathering requirements (Tag $\ni$ `web_search`) to Researcher agent.

4.2.2 Match software synthesis and execution (Tag $\ni$ `python, cli`) to Coder agent.

4.2.3 Match quality assurance and policy compliance (Tag $\ni$ `audit, verify`) to Validator agent.

4.3 Manage task lifecycle states in dispatcher queue:

4.3.1 Place sub-tasks in `PENDING` state until all parent dependency inputs resolve.

4.3.2 Transition verified nodes to `READY` and dispatch execution payloads to assigned agents.

Step 5: Agent Decision Cycle and Inter-Agent Communication

5.1 Execute autonomous Reasoning-Action (ReAct) cycle for each dispatched agent:

$$\text{State } s_t \longrightarrow \textbf{Observe} \longrightarrow \textbf{Reason} \longrightarrow \textbf{Act} \longrightarrow \textbf{Reflect}$$

Where: s_t: Current execution state at turn t | **Observe**: Ingest context & parent outputs $\mathcal{O}_i$ | **Reason**: Plan via CoT | **Act**: Invoke MCP tools | **Reflect**: Self-correct on error.

- **5.1.1** **Observe:** Ingest assigned sub-task description, operational constraints, and parent inputs.
- **5.1.2** **Reason:** Formulate intermediate chain-of-thought rationale to plan next tool invocation.
- **5.1.3** **Act:** Dispatch structured JSON-RPC tool calls to external MCP servers or formulate text.
- **5.1.4** **Reflect:** Inspect execution exit codes and output schemas; initiate local correction on errors.

5.2 Transmit intermediate artifacts via DAG edge message passing:

- **5.2.1** Route validated producer artifact $\mathcal{O}_i$ directly into consumer input buffer $\mathcal{I}_j$.
- **5.2.2** Enforce strict artifact isolation to prevent cross-task context contamination and token inflation.

5.3 Synchronize shared state via local blackboard memory:

- **5.3.1** Commit global project state, conversation turns, and file handles to IndexedDB (`PrismDB`).
- **5.3.2** Dispatch event-driven signals upon node resolution to awaken dependent downstream workers.

Step 6: Specialist Execution Workflows – Planner & Assistant

6.1 Execute Planner Agent workflow (System Architect & Coordinator):

- **6.1.1** Ingest user objective string and conversation context from `PrismDB`.
- **6.1.2** Decompose objective into atomic independent and dependent sub-goals.
- **6.1.3** Construct DAG graph vertices V, dependency edges E, and capability tags.
- **6.1.4** Solve provider routing optimization and enqueue topological schedule into Dispatcher.

6.2 Execute Assistant Agent workflow (Productivity & Communication):

- **6.2.1** Retrieve user communication preferences, profiles, and reference documents from storage.
- **6.2.2** Formulate structured response drafts, templates, or executive summaries.
- **6.2.3** Interface with Google Workspace MCP tools (Gmail draft creation, Drive document retrieval).
- **6.2.4** Emit verified communication draft; enforce approval gate prior to external message transmission.

Step 7: Specialist Execution Workflows – Researcher, Coder & Validator

7.1 Execute Researcher Agent workflow (Information Retrieval & Synthesis):

- **7.1.1** Formulate optimized web search queries and invoke Browser MCP server endpoints.
- **7.1.2** Scrape target web documents, parse DOM trees, and extract relevant contextual passages.
- **7.1.3** Deduplicate findings and compile structured Markdown report with verified source citations.

7.2 Execute Coder Agent workflow (Software Synthesis & Execution):

- **7.2.1** Inspect local repository tree via Filesystem MCP and synthesize target source code files.
- **7.2.2** Execute scripts inside Sandboxed Terminal CLI and inspect exit codes and error streams.
- **7.2.3** Analyze stack traces and compiler lints to autonomously self-correct syntax and runtime bugs.

7.3 Execute Validator Agent workflow (Quality Auditing & Safety Gate):

- **7.3.1** Scan candidate outputs against safety policies, token invariants, and functional constraints.
- **7.3.2** Emit binary verification signal: **PASS** to release artifact, or **FAIL** with error diagnostics.

Step 8: Standardized Tool Execution via Model Context Protocol

8.1 Interface agents to external tools via Model Context Protocol (MCP):

- **8.1.1** Standardize client-server communication using JSON-RPC protocol specifications.
- **8.1.2** Enable dynamic tool discovery: inspect exposed tool schemas, input parameters, and return types.
- **8.1.3** Decouple agent reasoning logic from external tool implementation details.

8.2 Orchestrate multi-domain tool execution ecosystem:

- **8.2.1** **System & Dev Tools:** Local Filesystem (file I/O), Sandboxed CLI (isolated shell), GitHub API.
- **8.2.2** **Productivity Tools:** Google Workspace MCP (Gmail messaging, Google Drive document access).
- **8.2.3** **Web Intelligence:** Browser MCP for real-time document search, retrieval, and scraping.

8.3 Enforce Human-in-the-Loop (HITL) authorization policy:

- **8.3.1** Permit autonomous execution for non-destructive, read-only tool queries.
- **8.3.2** Intercept state-mutating actions (file modification, email dispatch, CLI execution) for user approval.

Step 9: Context Compaction and Shared Memory Management

9.1 Address context window constraints and token consumption limits:

9.1.1 Mitigate fixed LLM context window boundaries to eliminate token overflow and latency degradation.

9.1.2 Prevent quadratic attention computation overhead across long multi-agent interaction sessions.

9.2 Execute dynamic context compaction and artifact pruning:

$$C_{t+1} = \mathrm{Prune}\big(C_t \cup \Delta\text{Artifacts} \setminus \text{Stale Tokens}\big)$$

Where: C_{t+1}, C_t: Next and current context buffers | ΔArtifacts: Verified upstream deliverables $\mathcal{O}_i$ | Stale Tokens: Discarded scratchpad tokens & raw tool logs | $\mathrm{Prune}(\cdot)$: Heuristic eviction filter.

9.2.1 Forward only verified parent output artifacts $\mathcal{O}_i$ to dependent downstream child nodes.

9.2.2 Evict intermediate reasoning scratchpads, redundant tool telemetry, and discarded draft tokens.

9.3 Persist global execution state using client-side database (`PrismDB`):

9.3.1 Commit session turns, task graphs, and tool execution logs to browser IndexedDB via Dexie.js.

9.3.2 Ensure client data privacy, full offline execution persistence, and instant crash recovery.

Step 10: Single-Stage Preference Alignment (ORPO) and Verification

10.1 Single-stage reference-free preference alignment formulation (ORPO):

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \lambda_{\text{OR}} \cdot \mathcal{L}_{\text{OR}} \quad \text{where} \quad \mathcal{L}_{\text{OR}} = -\log \sigma \left(\log \left[\frac{\text{odds}_\theta(y_w \mid x)}{\text{odds}_\theta(y_l \mid x)} \right] \right)$$

Where: $\mathcal{L}_{\text{ORPO}}$: Total loss | $\mathcal{L}_{\text{SFT}}$: SFT cross-entropy loss | λ_{OR}: Odds weight | $\mathcal{L}_{\text{OR}}$: Odds ratio penalty | $\text{odds}_\theta(y \mid x) = \frac{P_\theta(y|x)}{1-P_\theta(y|x)}$: Output sequence odds | y_w, y_l: Chosen / rejected responses for prompt x.

10.1.1 Jointly maximize chosen likelihood while penalizing rejected tokens ($2\times$ memory savings).

10.1.2 Supervise via UltraFeedback pairs $\mathcal{D} = \{(x, y_w, y_l)\}$, bypassing unstable RLHF loops.

10.2 Enforce runtime quality gating via preference score barrier $\mathcal{S}_{\text{ORPO}}(x, y) \geq \tau_{\text{approval}}$:

10.2.1 Audit deliverables using fine-tuned model (`reward_orpo_v2`, Qwen2.5-1.5B LoRA).

10.2.2 Score candidate outputs on instruction adherence, structural constraints, and safety guardrails.

10.3 Handle quality failures via targeted DAG re-planning:

10.3.1 Capture diagnostic error traces and schema deviations when output falls below τ_{approval}.

10.3.2 Dispatch error logs to DAG Planner for targeted branch re-planning without system abort.

Step 11: Artifact Assembly and Global State Commit

11.1 Aggregate and assemble multi-agent deliverables:

11.1.1 Harvest verified partial output artifacts $\mathcal{O}_i$ from all terminal nodes in the completed DAG.

11.1.2 Coalesce research summaries, verified code diffs, and formatted text into unified final deliverable.

11.2 Calculate end-to-end composite delivery quality metric:

$$\text{Delivery Quality } D = \text{Correctness} \times \text{Completeness} \times \text{Timeliness}$$

Where: $D \in [0, 1]$: Composite quality metric | Correctness: Schema conformance & execution pass rate | Completeness: Fulfillment of goal constraints | Timeliness: Latency penalty score.

11.2.1 Apply multiplicative scoring penalty: failure in any single dimension sharply penalizes overall quality.

11.2.2 Log quality vector to persistent performance ledger to calibrate future provider routing models.

11.3 Commit finalized session state with ACID guarantees:

11.3.1 Atomically write completed conversation records, DAG execution graphs, and tool receipts to `PrismDB`.

11.3.2 Record end-to-end telemetry: aggregate latency, per-agent token consumption, and audit receipts.

Step 12: Output Delivery and Reactive Workspace Presentation

12.1 Synthesize and package final deliverable artifacts:

- **12.1.1** Compile structured digital artifacts (Markdown reports, verified code patches, email/document drafts).
- **12.1.2** Generate comprehensive audit receipt containing agent provenance, execution duration, and tool logs.

12.2 Render presentation layer via multi-pane workspace interface:

- **12.2.1** Update interactive DAG HUD with real-time node states (`Queued` → `Running` → `Success` / `Failed`).
- **12.2.2** Mount Artifact Viewer canvas providing rich Markdown rendering, syntax highlighting, and export controls.
- **12.2.3** Present interactive approval cards for pending external actions with explicit authorization controls.

12.3 Synchronize client state reactively via local database transactions:

- **12.3.1** Commit verified state transitions and output artifacts directly into browser IndexedDB (`PrismDB`).
- **12.3.2** Trigger Dexie `useLiveQuery()` observers to update UI components reactively upon local database commits.

Thank You!

Questions & Discussion

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